

**Before you read — answer from memory**

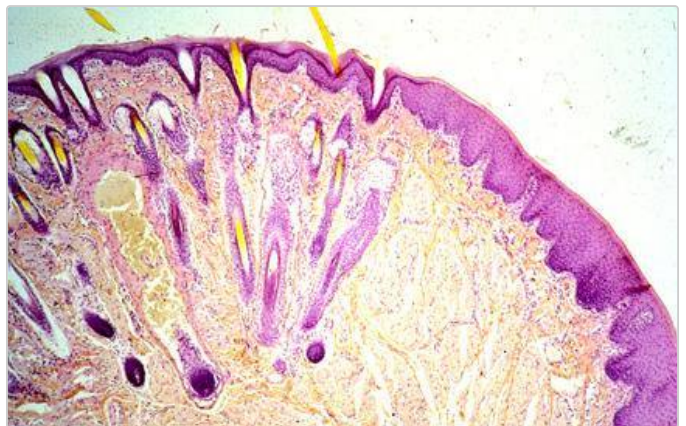
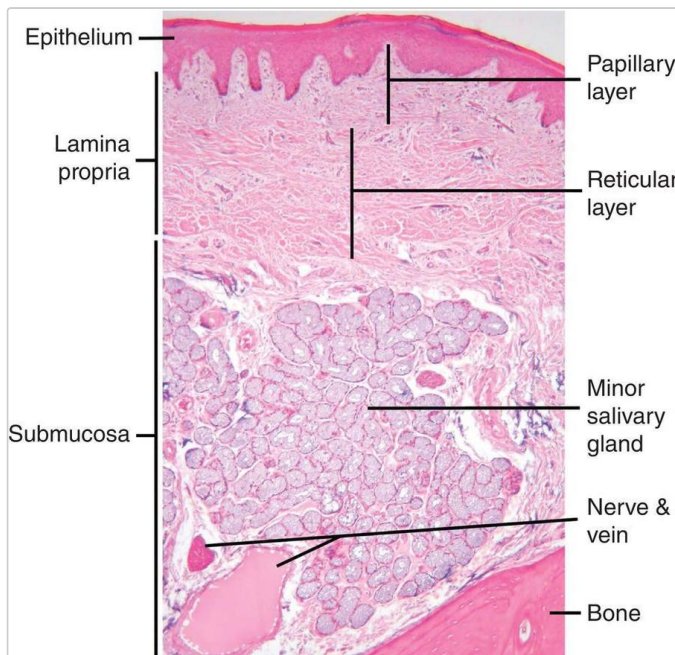
- A section of lip shows hair follicles and sweat glands. **Which surface** are you looking at?
- You see a papilla with taste buds on its **lateral wall**, not its tip. Which type — and which taste?
- Every acinus on the slide is dark and granular; not one pale tubule. **Name the gland.**

1 Oral cavity & lip

ID RULE — ORAL MUCOSA

Stratified squamous epithelium sitting on connective tissue, with no muscularis mucosae — then read what lies *beneath* to name the region.

Cue	Detail
Wall composition	Partly bone (jaws, hard palate) · partly muscle + connective tissue (lips, cheeks, soft palate, floor of mouth)
Lining	Mucous membrane = stratified squamous epithelium resting on connective tissue
Cheek	Connective tissue holds many elastic fibres and much fat
Hard palate	Mucous membrane rests directly on bone — the giveaway at low power
Soft palate	Core of skeletal muscle + numerous mucous glands in the submucosa
Deep lip	Orbicularis oris — striated bundles arranged concentrically around the oral orifice



Whole lip, low power. Outer skin (left) carries **hair follicles**; the vermilion (right) carries none. Learn the transition, not the picture.

Hard palate. Scan top → bottom: epithelium → lamina propria (papillary + reticular) → submucosa with **minor salivary glands** → bone. The bone at the base is the region-defining feature.

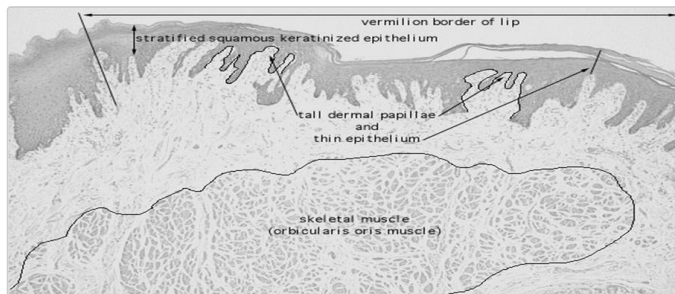
The three lip zones — the discriminator

Feature	Outer surface	Vermilion border	Inner surface
Epithelium	Thin skin	Lightly keratinized st. sq.	Thick non-keratinized st. sq.
Hair follicles	Present	Absent	Absent
Sweat glands	Present	Absent	Absent
Sebaceous glands	Present	Absent	Present

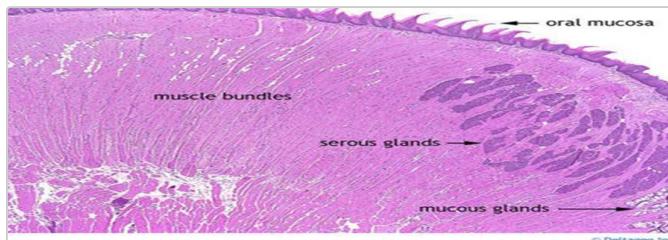
Other glands	—	—	Labial (minor salivary) glands
Vessels / nerves	Ordinary	Prominent vessels → red colour; numerous sensory endings	Ordinary

ONE FINDING THAT SEPARATES THEM

Hair follicles. Present → outer skin. Absent but epithelium still lightly keratinized with tall vascular papillae → vermillion. Absent and frankly non-keratinized with glands beneath → inner surface.



Vermilion border. Note **tall dermal papillae with thin overlying epithelium** — vessels ride close to the surface, which is why it is red. Skeletal muscle (orbicularis oris) fills the base.



Tongue, low power. Muscle bundles cut in several planes at once (the tri-orthogonal arrangement), with **serous and mucous glands** between them.

2 Tongue & papillae

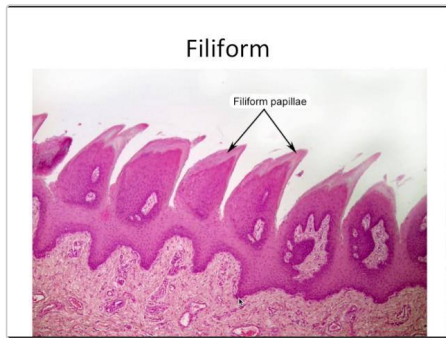
ID RULE – TONGUE

Skeletal muscle bundles cut in three planes in one field, capped by squamous epithelium bearing projections. No other organ shows muscle running three ways at once.

- A mass of **skeletal muscle**; occupies the floor of the mouth and fills the oral cavity
- Covered by **lightly keratinized** stratified squamous epithelium
- Mucosa bound **tightly** to muscle by the lamina propria — no loose submucosa to slide on
- Bundles of many sizes, separated by connective tissue, crossing in **3 planes** → flexibility for **speech, positioning food, chewing, swallowing**
- **Ventral** surface = floor of oral cavity → **non-keratinized** st. sq.
- **Dorsal** surface = **partly keratinized**, bearing papillae; each papilla = lining epithelium + **connective tissue core**
- Anterior **2/3** separated from posterior **1/3** by a **V-shaped groove** (sulcus terminalis)
- Behind the groove: **lingual tonsils**

Four papillae — discriminator grid

Papilla	Shape & size	Where	Taste buds	Key extra
Filiform	Smallest, most numerous ; elongated conical; partly keratinized	Entire dorsal surface	NONE	Gives roughness; provides friction for processing food
Fungiform	Mushroom shape; occurs singly ; epithelium slightly thinner than surrounding tongue	Evenly scattered between filiform	Scattered	CT core richly vascularised ; sweet & salty
Circumvallate	Largest, least numerous ; 8–12 in humans; ringed by a groove from infolded epithelium	V-shaped row at back of tongue	Numerous, on the LATERAL surfaces	von Ebner serous gland ducts empty into the groove; bitter
Foliate	Named in this lecture; not further described	—	Present	Listed among taste-bud-bearing papillae



Filiform. Pointed, conical, keratinized tips leaning in one direction. Nothing embedded in the epithelium — no taste buds.



Fungiform. Broad top on a narrower neck (mushroom), with a **wide vascular CT core** pushing up into it.



Circumvallate. The papilla sits **sunk below the surface**, ringed by a deep groove — the single most reliable low-power clue.

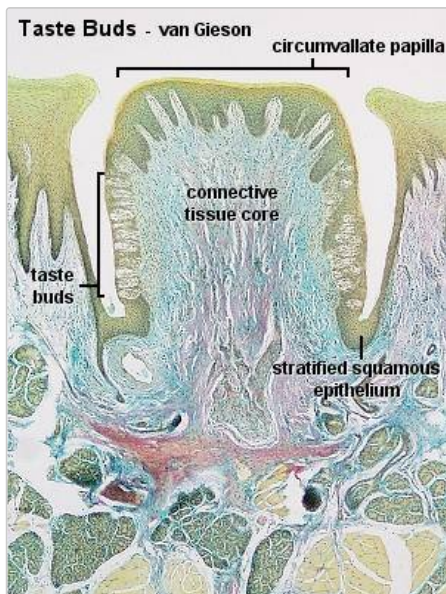
SCAN PATH FOR AN UNKNOWN PAPILLA

1. Does it project **above** the surface or sit **sunk in a groove**? → sunk = circumvallate.
2. If projecting: **pointed** (filiform) or **broad-topped** (fungiform)?
3. Only now go to $\times 40$ and look for **taste buds** as pale ovoid barrels in the epithelium.

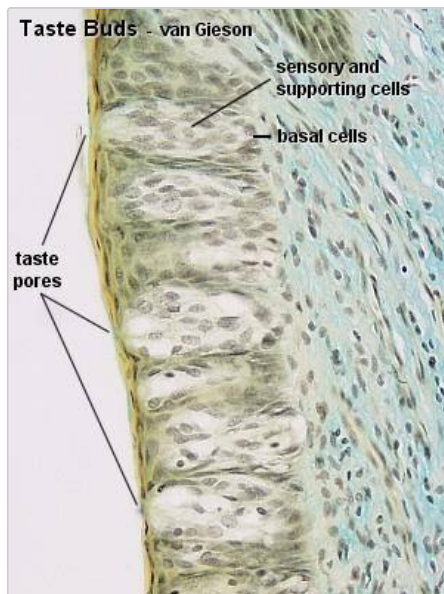
Taste buds

- Receptors for taste, located on the tongue in **circumvallate, fungiform and foliate** papillae
- In smaller numbers on the **soft palate, epiglottis, palatoglossal arches** and **posterior wall of the oropharynx**
- Cells seen: **sensory and supporting cells, basal cells**, opening at a **taste pore**

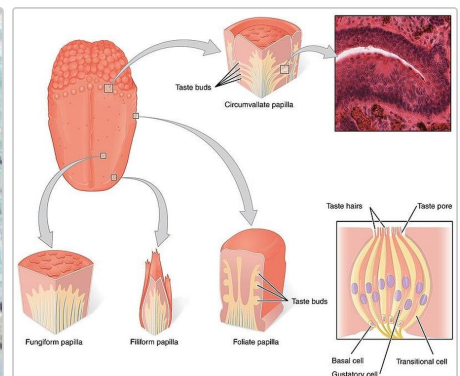
Taste	Best appreciated at
Sweet	Tip of the tongue
Salt	Just behind the tip and along the lateral border
Bitter	Circumvallate papillae



Taste buds in a circumvallate papilla. They line the **walls of the groove**, not the free upper surface — look sideways, not up.



Taste bud, high power. Pale barrel of cells spanning the epithelium, opening at a **taste pore**; basal cells sit at its foot.



The four papillae compared. Fungiform, filiform and foliate side by side, with the taste bud broken out: taste pore, taste hairs, gustatory, transitional and basal cells.

3 Salivary glands

- Two groups — **major**: parotid, submandibular, sublingual; **minor**: numerous small glands throughout the mouth
- Each major gland has a **capsule**; connective tissue **septa** divide it into **lobes** → **lobules**, carrying vessels and nerves — "like a branch of grapes"
- Every gland is built from just two things: **secretory cells** + a **duct system**

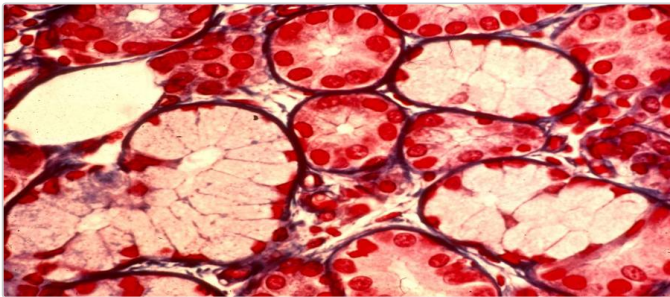
ID RULE – SALIVARY GLAND

Lobules of packed acini separated by CT septa, with ducts running inside the lobules. The colour of the acini then names the gland.

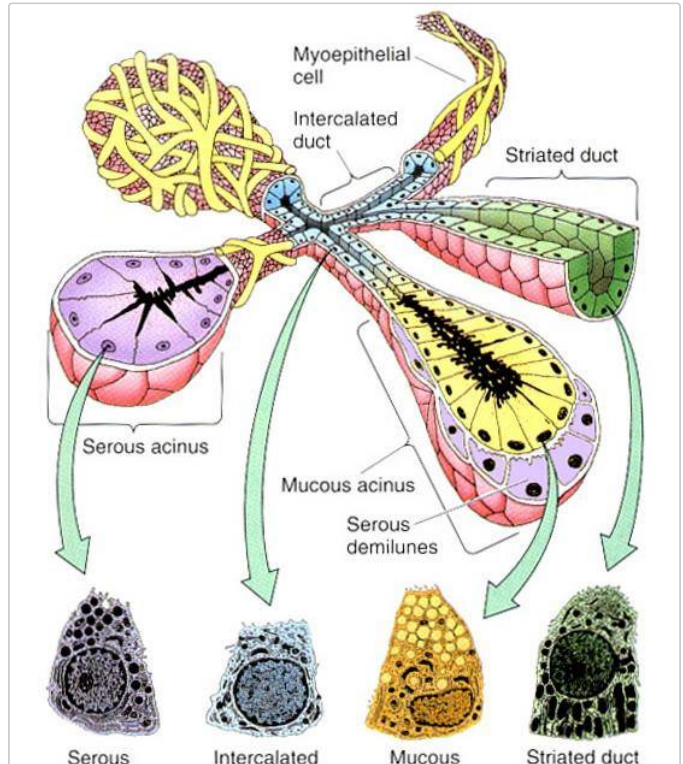
Serous vs mucous — the discriminator

Feature	Serous cell	Mucous cell
Shape	Pyramidal	Cuboidal to columnar
Arrangement	Spherical mass = acinus / alveolus , narrow lumen	Arranged as tubules
Cytoplasm	Basophilic , with protein secretory granules	Larger, more acidophilic than serous (per lecture)
Product	Protein, watery secretion	Mucous
Also present	Myoepithelial cells — the second (non-secretory) cell of the alveolus, squeezing secretion out	

Flag — check this against the lecturer. The slide describes mucous cells as "larger, more acidophilic than serous." In the lecture's own trichrome image the mucous units read as the **pale** ones and the serous as the dark ones. If a question turns on staining, go with what he said aloud.



Serous vs mucous. Dark, granular, small-lumen units = **serous**. Large, pale, foamy units with flattened basal nuclei = **mucous**. This one image is the whole discriminator.



Acinus → duct chain. Serous acinus, mucous acinus capped by **serous demilunes**, myoepithelial cells, then intercalated → striated duct.

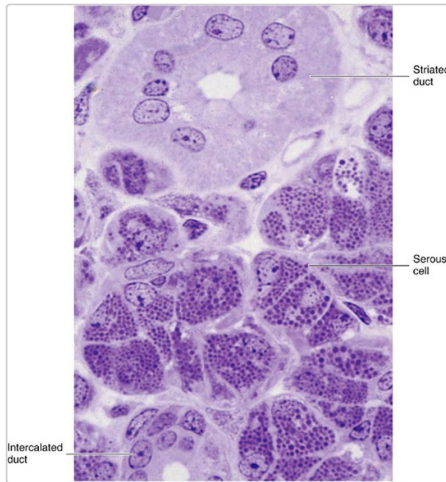
Duct system — in order of flow

Duct	Location	Epithelium	Recognise by
Intercalated	Intralobular — inside the lobule, thin CT sleeve	Simple cuboidal	Narrow lumen ; smallest calibre
Striated		Simple columnar	Basal striation — vertical lines in the basal cytoplasm
Interlobular	In the septa , between lobules and lobes	Simple columnar → stratified	Larger than intralobular; sits in a large amount of CT

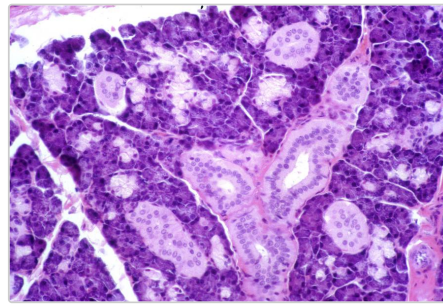
The three major glands

Gland	Type	Secretory cells	Share	Ducts	Opening
Parotid	Compound acinar ; largest gland; anterior to the ear	Serous only	25–30%	Long intercalated, short striated	One long duct, opposite the 2nd upper molar
Submandibular	Compound tubulo-acinar ; ovoid; below the mandible	Serous + mucous, serous predominant ; serous demilunes secrete lysozyme	70%	Long striated, short intercalated	Floor of mouth, either side of the frenulum
Sublingual	Compound tubulo-acinar ; smallest ; floor of mouth	Mucous + serous demilunes	5%	Short, ill-defined intercalated; striated ducts	10–12 small ducts

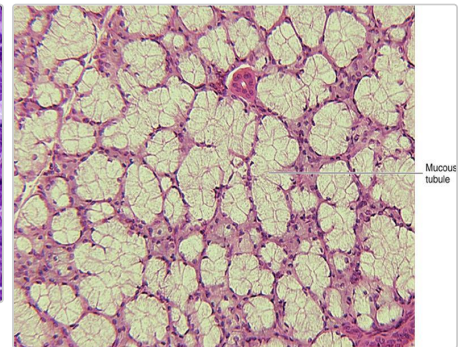
Flag — source arithmetic. The lecture's own figures (25–30% + 70% + 5%) total 100–105%. Quote them as given; just don't be caught out if a question asks them to sum.



Parotid. Uniformly dark serous acini throughout, with **intercalated** and **striated** ducts labelled. No pale tubules anywhere.



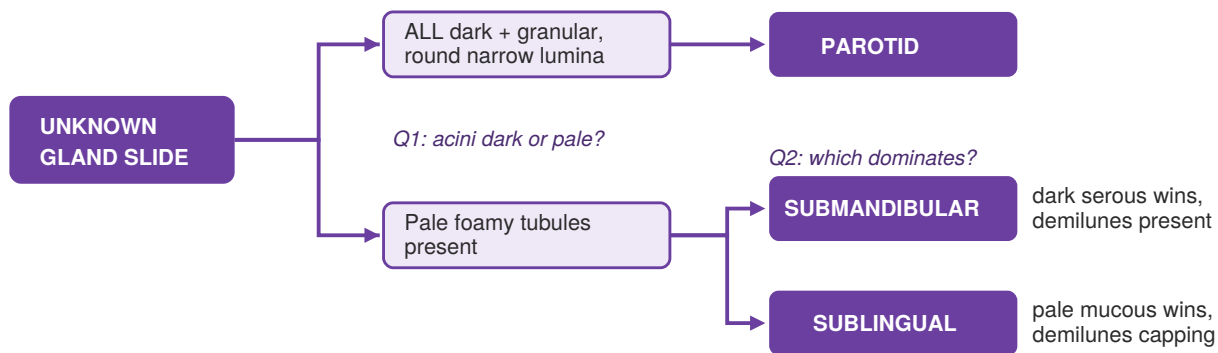
Submandibular. Dark serous predominates, but **striated ducts** are conspicuous and long — that prominence is itself the clue.



Sublingual. Field dominated by **pale mucous tubules**; serous appears only as small demilunes capping them.

Visual anchor — naming an unknown gland

SCAN PATH — low power first: 1 capsule → 2 septa & lobules → 3 acinus colour → 4 duct calibre. Only then go to ×40.



Pearls — if everything else goes, keep these

- **Vermilion** = no hair, no sweat, no sebaceous glands, but prominent vessels and lightly keratinized epithelium.
- **Filliform** has no taste buds. It is the only papilla type that doesn't.
- **Circumvallate**: taste buds on the lateral wall, von Ebner ducts into the groove, bitter.
- **Parotid** = pure serous, 25–30%. **Submandibular** = mixed, serous-predominant, 70%. **Sublingual** = mucous, 5%.
- **Striated duct** = basal striations + simple columnar + intralobular. Intercalated = simple cuboidal, narrow lumen.
- Parotid has **long intercalated** / **short striated**; submandibular is the **reverse**.

4 Test yourself — unseen fields

1. An unlabelled slide shows a mucosa with stratified squamous epithelium, tall connective-tissue papillae, prominent superficial vessels, and **no** hair follicles, sweat or sebaceous glands. Which region is this?
 - A. Inner surface of lip
 - B. Vermilion border
 - C. Outer surface of lip
 - D. Hard palate
2. A tongue section shows numerous small, pointed, conical projections covering the surface with no taste buds in any of them. What function does this papilla type serve?
 - A. Bitter taste detection
 - B. Sweet and salty detection
 - C. Friction during food processing
 - D. Draining von Ebner glands

3. A section shows a large papilla sunk below the surface, surrounded by a deep circular groove, with taste buds along its lateral walls and serous gland ducts emptying into the groove. Which taste modality is this papilla associated with?
- A. Sweet
 - B. Salt
 - C. Bitter
 - D. Sour
4. An unfamiliar salivary gland field shows uniformly dark, granular, pyramidal cells forming round acini with narrow lumina; no pale tubules are seen anywhere. Which gland is it, and what is the immediate corollary for its duct system?
- A. Sublingual; short ill-defined intercalated ducts
 - B. Parotid; long intercalated and short striated ducts
 - C. Submandibular; long striated and short intercalated ducts
 - D. A minor mucous gland; no striated ducts
5. A duct in a gland lobule is lined by simple columnar cells whose basal cytoplasm shows distinct vertical striations. Where is this duct located, and what is the next segment downstream?
- A. Intralobular; drains into an interlobular duct
 - B. Interlobular; drains into an intercalated duct
 - C. Intralobular; drains into an intercalated duct
 - D. Interlobular; drains into a striated duct

Answers — cover this block until you have committed to all five.

1. **B** — The vermillion is defined negatively as much as positively: lightly keratinized epithelium with prominent vessels (its red colour) and an absence of hair, sweat and sebaceous glands. Outer lip skin has all three; inner lip has labial and sebaceous glands.
2. **C** — Filiform papillae are the smallest and most numerous, partly keratinized, and contain no taste buds. They give the tongue roughness and provide friction that aids processing of food.
3. **C** — Circumvallate papillae, 8–12 in a V-shaped row at the back of the tongue, are surrounded by a groove formed by infolding of the epithelium. The ducts of the serous glands of von Ebner open into that groove, and bitter taste is attributed to them.
4. **B** — Purely serous acini identify the parotid. It is a compound acinar gland producing about 25–30% of saliva, with long intercalated ducts and short striated ducts — the reverse of the submandibular arrangement.
5. **A** — Basal striation plus simple columnar epithelium identifies a striated duct, which is intralobular. Flow runs acinus → intercalated (simple cuboidal, narrow lumen) → striated → interlobular duct in the septa, where the epithelium ranges from simple columnar to stratified.

Spaced review ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 21 ☐ Pre-exam